

Measures of Center

Wednesday, February 5, 2025 5:24 PM

In summary:

$$\text{mean } \bar{x} = 68.9$$

$$\text{median } \tilde{x} = 73.0$$

$$\text{mode } M = 73$$

which is Better?

Given a Quantitative (numeric) set of Data, one can ask the question: What is a typical value of the data?

For Example:

Suppose these are exam scores

{7, 64, 67, 70, 72, 73, 73, 73, 76, 82, 85, 85}

This question can be answered as follows:

I. The Mean - the average in which you add all the values and divide by the number of values.

$$\mu = 68.9 \quad \bar{x} = 68.9$$

population μ
("mean")

sample $\bar{x}$
("x-bar")

II. The Median - this is the center value in a set of sorted numbers. If an odd number of data, take the exact middle; if an even number of data, take midpoint of the two middle values.

$$\tilde{x} = 73.0$$

"x-tilde"

* NOTICE
ROUNDING

III. The Mode - this is the most frequently occurring value; sometimes there might be two modes and this is called bi-modal. If more than two modes, it is then called multi-modal; however, that has no practical value.

$$M = 73$$

no rounding

Note: throughout this textbook, measures of center are referred to as ~~Typical Values~~; however, that is NOT the terminology used in Statistics. These values are called Measures of Center.

Measures of Center tell us approximately where the center of the data is.

I. The Mean

Tuesday, October 22, 2024

1:50 PM

scratch

$$59/4 = 14.75$$

FINDING THE MEAN

A **typical value** is an ^{datum} amount that represents the values in a set of numbers. A typical value must fall within the series of numbers it represents. There are three common typical values in a set of numbers: **mean**, **median**, and **mode**.
1. 2. 3.

When most people use the word **average**, they are talking about the **mean**. For a set of values that are similar to each other, the mean is what you want to know when you ask, "About how much . . .?"

To find the mean, add the values in a set of data, and then divide the sum by the number of values in the set.

FINDING THE MEAN

Example Ella scored 14, 18, 11, and 16 points in her last 4 basketball games. What is her scoring average?

Step 1

Find the mean. Add the values in the set.

$$14 + 18 + 11 + 16 = 59$$

Ella's scoring average is **14.75 points**.

Step 2

Divide the sum by the number of values in the set.

$$\frac{59}{4} = 14\frac{3}{4} = 14.75$$

The mean is not necessarily equal to any number in the set. However, it must fall within the range of the numbers in the set.

$$\{11, 14, 16, 18\} \quad \mu = 14.8$$

Population, Not a Random Sample

Examples of the Mean

Wednesday, January 29, 2025

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When calculating the mean (average), always write one more decimal place for the mean. For example, if the data are whole numbers, write the mean with one decimal point. The only time you would break this rule is if the extra decimal makes no sense; e.g., a mean of \$1.583 makes no sense! Also, when rounding, if 5 or more, round up; if less than 5, round down.

ASSUME THESE ARE ALL SAMPLES.

Find the mean for each set of numbers.

1. 35, 49, 31, 25

$$\bar{x} = 35.0$$

\$1.25, \$2.08, \$0.97, \$3.12, \$1.48

$$\bar{x} = \$1.78$$

common sense.
rounding

5, 4, 12

$$\bar{x} = 7.0$$

2. 174, 12, 398, 214, 45

$$\bar{x} = 168.6$$

6, 3, 15, 9, 4

$$\bar{x} = 7.4$$

\$35, \$12, \$97, \$6, \$51, \$90

$$\bar{x} = 48.50$$

3. \$215, \$154, \$192

$$\bar{x} = \$187$$

87, 96, 95, 100, 75, 82, 88

$$\bar{x} = 89.0$$

32.1, 15.75, 17.2, 18.35, 28.4

$$\bar{x} = 22.36$$

4. 5.6, 5.1, 6.4

$$\bar{x} = 5.70$$

\$240, \$198, \$264, \$249, \$220

$$\bar{x} = \$234.20$$

1, 7, 13, 7, 10

$$\bar{x} = 7.6$$

Solve.

5. In July, DeShaun's weekly grocery bills were \$84, \$98, \$72, and \$60. What is his mean weekly grocery bill for the month?

NOT A RANDOM SAMPLE

$$\mu = \$78.50$$

Pop: all weeks in a particular month

6. Miranda received scores of 83, 78, 85, 82, and 77 on 5 tests. What is her average score on the 5 tests?

Not sure if these are all her tests μ or just a sample $\bar{x}$

$$81.0$$

Examples of the Mean, cont'd...

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If you know the mean, you can use algebra to find a missing value. You can also use the mean to draw conclusions and make predictions.

There are called:

CONFIDENCE INTERVALS
& HYPOTHESIS TESTING

USING THE MEAN

Example 1 Sean budgets \$400 a month for food. In 5 months, he spent \$415, \$380, \$489, \$392, and \$404. In order to average \$400 per month, how much should he spend in the sixth month for food?

Step 1

Write an equation to represent the mean. Let x represent the amount spent in the sixth month.

$$\frac{415 + 380 + 489 + 392 + 404 + x}{6} = 400$$

Sean should spend **\$320** in the sixth month to average \$400 per month.

Example 2 Gina is paid weekly based on the amount of sales she makes. Her first paychecks are for the following amounts: \$346, \$358, \$320, and \$336. Based on the mean for the first 4 weeks, about how much will Gina make in a year?

Step 1

Find the mean of the first 4 weeks.

$$\frac{346 + 358 + 320 + 336}{4} = 340$$

$$\mu = \$340$$

Gina will make about **\$17,680** in a year.

Step 2

Solve for x .

$$\frac{415 + 380 + 489 + 392 + 404 + x}{6} = 400$$

$$\frac{2,080 + x}{6} = 400$$

$$2,080 + x = 2,400$$

$$x = 320$$

special note

if there are n data

then there are $n-1$

Degrees of Freedom

$\therefore n=6$ } means
 $df=5$ } there
is only
one possible
answer for x .

Step 2

Gina will typically make \$340 a week. There are 52 weeks in a year. Multiply the mean by 52.

$$\$340 \times 52 = \$17,680$$

yearly income \$17,680

Examples of the Mean, cont'd...2...

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Solve as directed.

7. Jena plans to run an average of 6 miles a day. For the first 6 days, she runs 3, 8, 8, 0, 4, and 5 miles. How many miles should she run on the seventh day to meet her goal?

$$\mu = 6.0$$

$$\frac{3+8+8+0+4+5+x}{7} = 6$$

$$x = 14 \text{ miles}$$

8. In the first 8 weeks of school, students at Pine Elementary read the following numbers of books.

80 85 92 78 80 81 90 96

Based on the average of the first 8 weeks, how many books will they read in a 36-week school year?

$$\mu = 85.25$$

$$= 85.3 \text{ per wk}$$

$$36 \text{ wks} (85.3) = 3070.8$$
$$= 3071$$

9. Jared set a goal to sell 300 alarm systems in a 60-day period. At the end of the 60 days, he had averaged 4 alarms per day. Did he meet his goal?

$$60 \times 4 = 240$$

NO.

10. **Explain** Carmel received test scores of 77, 81, 95, and 91. She has 1 more test to take. In order to get an A in the class, she needs an average of 90 on the 5 tests. If the highest score Carmel can earn on the last test is 100, is it possible for her to get an A in the class? Explain your thinking.

$$\frac{77+81+95+91+x}{5} = 90$$

$$x = 450 - 344$$

$$x = 106$$

NOT possible!!

II. The Median

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MEDIAN

There are other typical values that might be more representative of a set of data. The median is the middle value of a set of data. To find the median, sort the data from least to greatest. For an odd number of values, the middle number is the median. For an even number of values, the median is the mean of the two middle numbers.

FINDING THE MEDIAN

{ 10, 30, 35, 40, 140 }

Example Angela has helped 5 dogs onto the examination table. Their weights are 40 lb, 10 lb, 140 lb, 35 lb, and 30 lb. What is the median weight of the dogs?

$$\bar{x} = 35.0 \text{ lb}$$

Step 1

Sort the numbers numerically from least to greatest.

10 lb 30 lb 35 lb 40 lb 140 lb.

The median weight is **35 lb**.

Step 2

Choose the middle number from the set. (If there is an even number of values, find the mean of the two middle values.)

Find the median.

1. \$12, \$75, \$4, \$23, \$51

$$\{4, 12, 23, 51, 75\}$$
$$\bar{x} = 23$$

53, 22, 12, 18, 50

$$\{12, 18, 22, 50, 53\}$$
$$\bar{x} = 22.0$$

13.4, 8.6, 4.1, 80.9

$$\{4.1, 8.6, 13.4, 80.9\}$$
$$\bar{x} = (8.6 + 13.4) / 2$$
$$= 11.00$$

2. 0.8, 1.4, 2.3, 0.6, 1.8

$$\bar{x} = 1.40$$

62, 78, 73, 45, 67, 90

$$\bar{x} = \frac{73 + 67}{2}$$

$$\bar{x} = 70.0$$

\$457, \$375, \$223, \$480

$$\bar{x} = (457 + 375) / 2$$

$$\bar{x} = \$416$$

* money - common sense rounding

check

$$\{0.6, 0.8, 1.4, 1.8\}$$
$$2.3$$

III. The Mode

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MODE

Measure of Center

Another typical value is the ~~mode~~ **mode**—the value in the set that occurs most often. (If each value occurs only once, there is no mode in the set.)

FINDING THE MODE

Example A restaurant offers these prices on complete meals: Fried Chicken—\$8.99, Steak—\$10.99, Fish—\$8.99, Shrimp—\$10.99, Lasagna—\$8.99, Hamburger—\$7.99. What is the mode for these prices?

$$m = \$8.99$$

Count how many times each value occurs.

\$7.99: 1 time \$8.99: 3 times \$10.99: 2 times

Because it occurs the most (3 times), **\$8.99** is the mode.

Find the mode.

3. 35, 18, 21, 21, 18, 97, 32, 18

$$m = 18$$

22°, 17°, 31°, 22°, 18°, 35°, 22°, 31°

$$m = 22^\circ$$

4. \$6.50, \$5.75, \$6.00, \$5.50, \$6.75

No Mode

23, 24, 26, 24, 28, 23, 25, 23, 27, 25, 24

$$\begin{array}{l} 23 \quad 3\text{'s} \\ 24 \quad 3\text{'s} \end{array}$$

Bimodal

$$m = 23 \text{ \& } 24$$

CONSIDER A PRIMARY SCHOOL



AGES OF
EVERYONE IN
THE SCHOOL

WHY??

Outliers

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note: data (plural)
datum (singular)

An outlier is a datum that is significantly far from the other data.

For example, consider the data set {23, 69, 73, 76, 77, 82, 85, 88, 89, 91, 94, 99}.

Which datum is a far from the others? 23

Therefore, 23 is an outlier.

CALCULATE : mean $\mu = 78.8$

Median $\tilde{x} = 83.5$

Trimmed mean $\mu = 83.9$

Comparing the Mean & the Median

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Sometimes, one or more data values are very large or very small compared to the other values in the set. These extreme values are called outliers. Outliers don't affect the median as much as they affect the mean.

*hardly at all !!
(usually)*

USING THE MEAN AND MEDIAN WITH OUTLIERS

Example A jewelry store has 5 different rings on sale for \$27, \$26, \$33, \$31, and \$108.

Your common sense tells you that \$30 is a typical price for a ring. Compare how the outlier of \$108 impacts the mean more than the median.

Find the trimmed mean $\mu = \$2925$

Find the mean of the 5 rings:

$$\begin{aligned} \$27 + \$26 + \$33 + \$31 + \$108 &= \$225 \\ \$225 \div 5 &= \$45 \end{aligned}$$

$$\mu = \$45$$

The mean, \$45, is much higher than \$30.

Find the median of the 5 rings:

$$\begin{array}{ccccc} \$26 & \$27 & \$31 & \$33 & \$108 \\ \text{The middle value is} & \text{\$31.} \end{array}$$

$$\tilde{x} = \$31$$

The median, \$31, is very close to \$30.

Notice in the example above that while the mean may not be a typical value, the median is.

a good measure of center

First, identify the outlier. Then find the mean and the median of the data set.

5. 13.4 kg, 14.2 kg, 3.2 kg, and 12.8 kg

Outlier: 3.2

Mean: $\mu = 10.90$

Median: $\tilde{x} = 13.10$

$$\sigma = 4.47$$

$$\begin{aligned} \mu \pm 2\sigma &= 10.90 \pm 2(4.47) \\ &= 1.96 \text{ to } 19.84 \end{aligned}$$

7. 61 in, 68 in, 44 in, 60 in, 69 in, and 64 in

Outlier: 44

Mean: $\mu = 61.0$

Median: $\tilde{x} = 62.5$

$$\sigma = 8.3$$

$$\begin{aligned} 61.0 \pm 2(8.3) \\ = 44.4 \text{ to } 77.6 \end{aligned}$$

6. \$126,000; \$74,000; \$86,000; \$85,000; and \$84,000

Outlier: \$126K

Mean: $\mu = \$91K$

Median: $\tilde{x} = \$85K$

$$\sigma = 18$$

$$\begin{aligned} \mu \pm 2\sigma &= 91 \pm 2(18) \\ &= 55 \text{ to } 127 \end{aligned}$$

8. 73, 69, 76, 107, 80

Outlier: 107

Mean: $\mu = 81.0$

Median: $\tilde{x} = 76.0$

$$\sigma = 13.5$$

$$\begin{aligned} \mu \pm 2\sigma &= 81.0 \pm 2(13.5) \\ &= 54 \text{ to } 108 \end{aligned}$$

$$= 55 \text{ to } 127$$

$$= 54 \text{ to } 108$$

Solve.

9. The following data shows the number of sales made in a month by 8 out of 9 salespeople.

45 34 31 48 37 43 32 42

The 9th employee made 66 sales in that month. By how much does the mean change when the 9th value is added?

$$\mu_8 = 39.0 \quad \bar{x}_8 = 39.5$$

$$\mu_9 = 42.0 \quad \bar{x}_9 = 42.0$$

10. **Explain** The data below gives the margin of victory for a high school football team's games in a season.

~~3~~ 7 ~~21~~ ~~14~~ ~~3~~ ~~17~~ ~~30~~ ~~3~~

Do you think the mean, median, or mode is the most reliable typical value for the data? Explain

$$\mu = 12.3$$

$$\bar{x} = 10.5$$

$$M = 3$$

Measures of Center

USING TYPICAL VALUES

When news sources report the results of a survey or study, they don't usually tell you all of the data. Instead, they might sum up the study with a typical value. You must use your knowledge of mean, median, and mode to interpret the information.

with measures of center and/or a graph

Here are some useful tips to remember:

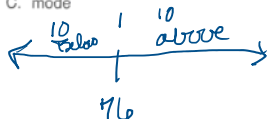
- ✓ When people say "average," they're usually talking about the mean. Be careful: outliers can make the mean misleading.
- ✓ The median is the middle value. Outliers don't have a big effect on the median.
- ✓ The mode is the value that occurs most often. The mode is useful when you care what is popular or what happens frequently.

Quantitative data only!!

Use your knowledge of mean, median, and mode to answer the following questions.

1. In Mr. Bound's class, 1 student scored a 76 on a test, 10 students scored above 76, and 10 students scored below 76. What typical value does a score of 76 represent?

- A. mean
B. median
C. mode



Because the median is the physical middle of SORTED numeric data

2. The manager at a toy store wants to reorder the items that sell the best. Which typical value should he use to decide which items to reorder?

- A. mean
B. median
C. mode

Because TOYS are NOT quantitative & cannot find the "average" toy.

consider:

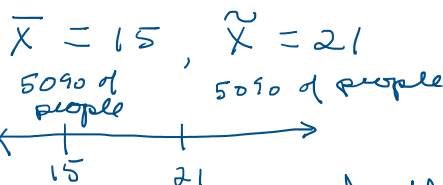
TOY	QTY
DOLL	175
TRUNK	158
GAME	183

DATA

COUNTS OF DATA BUT NOT DATA

4. An office poll asks employees how many cups of coffee they drink in a week. The mean of the data is 15, but the median is 21. Which of the following is true?

- A. A few people drink much fewer than 21 cups of coffee a week.
B. A few people drink much more than 21 cups of coffee a week.
C. Most people drink between 15 and 21 cups of coffee a week.



the mean is pulled left, so some outliers.

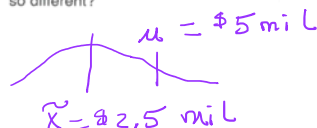
5. Out of the 25 students in a 5th grade class, 22 are between 49 and 51 inches tall. Three students are 59 inches tall. Which of the following must be true?

- A. The median is higher than the mean.
B. The mean is higher than the median.

height	QTY
49"-51"	22
59"	3

outliers pull mean up

6. Explain The mean salary for NBA basketball players is about \$5 million a year. The median salary for NBA players is about \$2.5 million a year. Why do you think the two typical values are so different?



There are a few highly paid basketball superstars.

∴ Mean significantly lower than median?

mean median

DATA

3. **Explain** You are researching housing prices. You can afford a house that will sell for about \$150,000. You know the mean, median, and mode for a certain neighborhood. Which typical value would be most helpful in deciding whether you can afford a house in this neighborhood? Explain your answer.

Mode? No, There could be 10 houses @ \$150K but the rest in the millions

Mean? No, a few high outliers could pull the mean high making it appear it's an expensive area.

Median? Yes!! If $x = \$175K$, you know 50% of houses will be less than that.

∴ ① Mean significantly lower than Median?

Low (Left) outliers

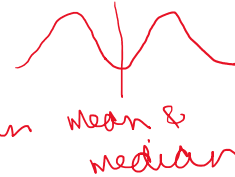
② Mean significantly higher than Median?

high (Right) outliers

③ Mean & Median close?

No outliers

Data might be symmetric.



Graphing Typical Values

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Sometimes line graphs display the mean, median, or mode of a data set. These graphs can show how typical values change over time.

measures of center

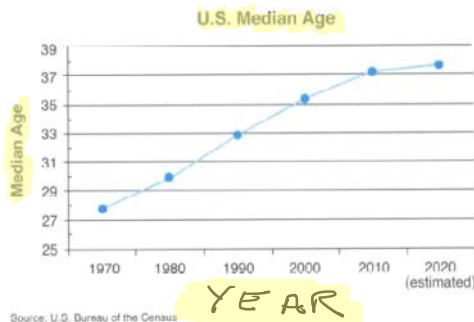
But Be Careful...
an average of averages
used, has no meaning!!

USING TYPICAL VALUES TO INTERPRET GRAPHS

Example This graph shows the U.S. median age from 1970 to the estimated value for 2020. What can you conclude about the U.S. population?

You can see from the graph that the U.S. median age has increased from 1970 to the present. But what does the median say about the whole population?

The median is the middle value. In 2010, the median age was 37. This means that the number of people younger than 37 was equal to the number of people older than 37. You can imagine there are many people younger than 37, so there must be many people older than 37 as well.



If the median age in the United States is rising, then you can conclude that the population as a whole is getting older.

∴ the population is
aging w/ fewer
Births.

This is what happened with Mao's
absurd 1-child policy!! Now, Xi is
trying to reverse the trend. Man's
wisdom (Mao) vs. God's wisdom (be
fruitful and multiply as written in
Genesis 1:28).

Graph Examples

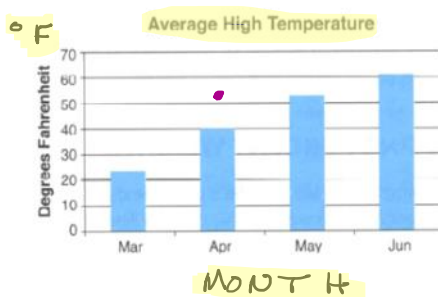
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Use the bar graph for problems 7 and 8.

The graph shows the average high temperature in a forest from March to June. Researchers recorded the high temperature for each day in a month and calculated the average of those values.

7. Which typical value does this graph use?

The mean



8. On April 28, the high temperature was 51°F. What can you conclude about other days in April?

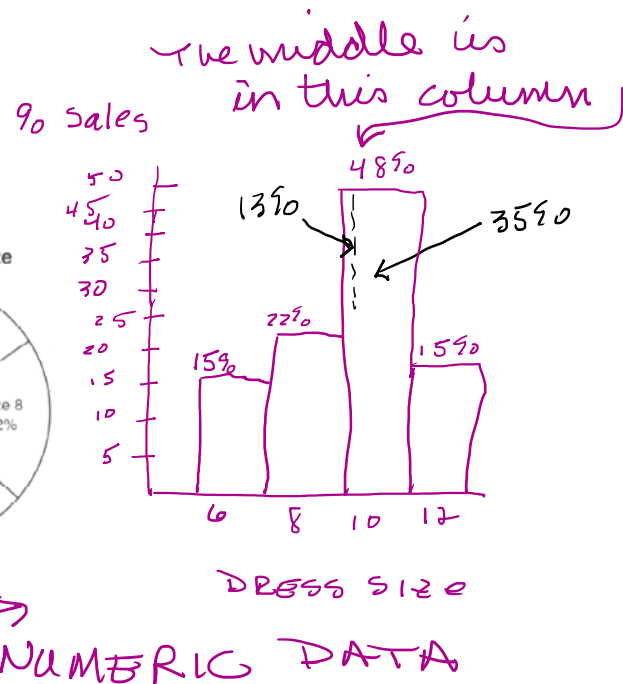
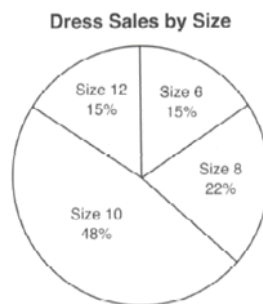
51° F was an outlier

Use the circle graph for problems 9 and 10.

The graph shows the breakdown of dress sales at a boutique over one weekend.

9. Which size represents the mode of the data?

Size 10 @ 48%



10. Explain Can you determine from the graph what the median size would be? Explain your answer.

Size 10.. See histogram

NUMERIC DATA

WEIGHTED AVERAGES

Sometimes, the values in a data set may not be equally important. For example, a final exam might count as two tests. If you were finding your average in the class, your final exam score would be worth more than your other scores.

When some values in a data set are worth more than others, you must find the weighted average.

FINDING A WEIGHTED AVERAGE

Example Aparna's average for the first 4 tests is 80. Her score on the fifth test is 90. What is her average in the class?

$$(4/5)(80) + (1/5)(90) = 82$$

You might be tempted to say 85 because it is the mean of 80 and 90. Look at the diagram to the right. Note that 80 represents 4 tests, but 90 represents only 1. You must find the weighted average.



Step 1

Multiply each score by the weight. Then add.

$$(80 \times 4) + (90 \times 1) = 410$$

Aparna's average is 82.

Step 2

Divide by the number of values. Aparna took 5 tests.

$$410 \div 5 = 82$$

Some averages are calculated by assigning percentages to different categories. These percentages must add up to 100%. Since the percentages represent one whole, you don't need to divide by the number of values.

USING PERCENT WEIGHTS

Example In Ms. Padilla's class, tests are worth 30% of the final grade, homework is worth 20%, participation is worth 10%, and the final exam is worth 40%. Jeff earns an 80 on the tests, 95 on homework, 100 on participation, and an 85 on the final exam. What is his average in the class?

Step 1

Multiply Jeff's scores by the percent weights.

Tests: $80 \times 0.3 = 24$

Homework: $95 \times 0.2 = 19$

Participation: $100 \times 0.1 = 10$

Final Exam: $85 \times 0.4 = 34$

Jeff has an average of 87 in the class.

Step 2

Add.

$$24 + 19 + 10 + 34 = 87$$

EASY - PEASY

$$\begin{array}{rcl} \text{Tests } 30\% & \times & 80 = 24 \\ \text{Hw } 20\% & \times & 95 = 19 \\ \text{part. } 10\% & \times & 100 = 10 \\ \text{Exam } 40\% & \times & 85 = 34 \\ \hline & & 87 \end{array}$$

SYLLABUS

Hw 20% of course grade

QZ 30%

EXAM 50%

Your scores

Hw 88

QZ 79

exam 63

what is your course grade?

score	weight	pts
88	.20	17.6
79	.30	23.7
63	.50	31.5
		<u>72.8</u>

you passed !!

Weighted Averages Examples

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Solve as directed.

1. On a recent road trip, Cesar averaged 65 mph for the first 4 hours and 80 mph for the final 6 hours. What was his average speed for the entire trip?

$$4 \text{ hrs} + 6 \text{ hrs} = 10 \text{ hrs}$$

$$(4/10)(65) + (6/10)(80) = 74 \text{ mph}$$

2. An online food critic rates restaurants based on four criteria. She assigns the following weights to the categories.

Category	Weight
Taste	8 x 35% = 2.8
Presentation	7.8 x 20% = 1.56
Service	8.4 x 25% = 2.1
Atmosphere	8.2 x 20% = 1.64

Blossom Café receives the following: 8 for Taste, 7.8 for Presentation, 8.4 for Service, and 8.2 for Atmosphere. What is Blossom Café's total weighted score?

8.1

3. The table shows how many hours Yeri worked each day during the past three months.

Number of Hours	Number of Days
3	3
6	18
8	54
11	15

What is the average number of hours Yeri worked per day over the past three months? Round your answer to the nearest tenth. (Hint: Treat the number of days as the weight.)

$$7.9 \text{ hrs/day}$$

TOTAL HRS

$$\begin{array}{r} 9 \\ 108 \\ 432 \\ 165 \end{array}$$

714 hrs in 90 DAYS

$$714 \div 90 = 7.9 \bar{3}$$

4. In the first 10 games of the season, a quarterback has a 60% pass completion rate. In the last 5 games, his pass completion rate is 78%. What is his pass completion rate for the entire season?

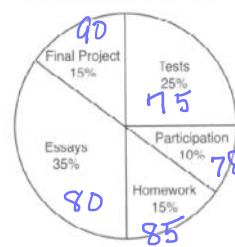
$$\begin{aligned} & (10/15)(.6) + (5/15)(.78) \\ & = 2/3(4/10) + (1/3)(78/100) \\ & = 4/10 + 26/100 \\ & = 40/100 + 26/100 \\ & = 66/100 \\ & = 66\% \end{aligned}$$

5. At a gymnastics meet, Michelle receives 152 points for execution and 110 points for artistry. Execution makes up 65% of her total score, while artistry makes up 35% of her score. What is Michelle's final score?

$$152(.65) + 110(.35) = 137.3$$

6. Mr. Reyes gives his students the following circle graph to explain their final grade.

Final Grade Breakdown



$$\begin{aligned} (.15)(90) &= 13.5 \\ (.25)(75) &= 18.75 \\ (.10)(78) &= 7.8 \\ (.15)(85) &= 12.75 \\ (.35)(80) &= 28 \\ \hline &80.8 \end{aligned}$$

Christian receives a 75 for tests, a 78 for participation, an 85 for homework, an 80 for essays, and a 90 for the final project. What is his final grade in the class, rounded to the nearest whole number?

81

7. Paul buys two suits for \$175 each at one store and three suits for \$150 each at a different store. How much did Paul pay on average per suit?

$$\begin{aligned} & 5 \text{ suits} \\ & (2/5)(175) + (3/5)(150) \\ & = \$160 \end{aligned}$$

8. After the first three months of the season, a baseball player's batting average is .300. After another month, his average for the four months is .275. Which of the following is most likely?

- A. He did not play the last month.
B. He batted about .200 the last month.
C. He batted about .300 the last month.
D. He batted about .350 the last month.

$$\begin{aligned} & \text{month \# 1} \text{ — } .3 \\ & \text{2 — } .3 \\ & \text{3 — } .4 \\ & \text{4 — } x \\ & .3 + .3 + .3 + x = 1.1 \\ & 1.1 = .275 \times 4 \\ & 1.1 = 1.1 \\ & x = .2 \end{aligned}$$